

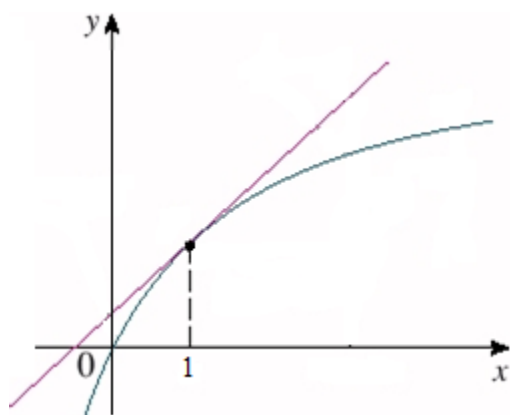
4.2 Linear Approximation & Differentials

In this section we will go over another important application of differentiation, and we will attach some rigorous meaning to the notion of a differential.

Motivation: What is an example of a function that is easy to calculate for some inputs, but quite difficult to compute by hand for other inputs?

There are many possible answers to this question. For example trigonometric functions are easy to compute at angle measures that typically get drawn on the unit circle but difficult elsewhere. Another example of such a function is $f(x) = \sqrt{x}$. This function is easy to compute at the perfect squares but difficult to compute to a high level of accuracy for real numbers that are not squares of rational numbers.

Suppose for example we would like to try to approximate the exact value $\sqrt{\frac{3}{2}}$ without using a computational device. To do this we notice that it would be easy for us to find the tangent line to the curve determined by f at $(1, f(1))$ since the square root of one is easy to compute. We illustrate this idea with the following picture:



Note that if $x = 1$ then $f(1) = \sqrt{1} = 1$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so

$$m = f'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2}.$$

Therefore the equation of this tangent line is:

$$y - 1 = \frac{1}{2}(x - 1) \Leftrightarrow y = \frac{1}{2}x + \frac{1}{2}.$$

Notice that the tangent line is really close to the actual curve for x -values near 1, that is, $f(x) \approx y(x)$ for x -values near 1.

So, to approximate $\sqrt{\frac{3}{2}}$ we could just evaluate the tangent line at $x = \frac{3}{2}$.

Therefore we can say: $\sqrt{\frac{3}{2}} = f\left(\frac{3}{2}\right) \approx y\left(\frac{3}{2}\right) = \frac{1}{2}\left(\frac{3}{2}\right) + \frac{1}{2} = 1.25$.

Using the calculator we see that $\sqrt{\frac{3}{2}} \approx 1.224744871$. So, our method of using the tangent line got us one decimal place of accuracy.

Note: An important question is: How can we tell how far off an approximation with a tangent line will be from the actual value we seek with no computation device? This question is very important in computer science and will be addressed in Calculus II. In Calculus I we are in the unfortunate position of knowing that the tangent line yields approximations, but the only way we can tell how good an approximation is, is to use the calculator. The discussion in the motivation leads to the following definition.

Definition: The line $L(x) = f'(a)(x-a) + f(a)$ is called the **linearization** of f at a .

The approximation $f(x) \approx L(x)$ is called the **linear approximation** of f near a .

Note: In the motivation, we found that the linearization of $f(x) = \sqrt{x}$ at 1 was $L(x) = \frac{1}{2}x + \frac{1}{2}$.

Example 1: Consider the function $f(x) = x^6$.

- a) Find the linearization of f at $x = 2$.

Note that if $x = 2$ then $f(2) = (2)^6 = 64$. So we are looking for the tangent line to f at the point $(2, 64)$.

Also, $f'(x) = 6x^5$, so $f'(2) = 6(2)^5 = 192$.

Therefore the linearization is: $L(x) - 64 = 192(x - 2) \Leftrightarrow L(x) = 192(x - 2) + 64$

- b) Use your answer from part (a) to approximate the number $(2.01)^6$ by hand.

Using the linear approximation of f near 2, we have that:

$$(2.01)^6 = f(2.01) \approx L(2.01) = 192(2.01 - 2) + 64 = 192(0.01) + 64 = 1.92 + 64 = 65.92$$

- c) Use your calculator to find how far off your approximation in part (b) is from the actual value of $(2.01)^6$

Using the calculator, we see that $(2.01)^6 \approx 65.9441606$, which is larger than what we estimated to be.

Our approximation was off by $2.01^6 - 65.92 \approx .0241606012$

Question: We notice that in the motivation the linearization gave us an over approximation of the actual value. In Example 1 the linearization gave us an under approximation. Why is this?

Example 2: Use a linear approximation to estimate the value of $\ln(1.5)$.

Let $f(x) = \ln(x)$ We will find a linearization of f at $x = 1$, since $f(1)$ is easy to compute and 1 is close to 1.5.

If $x = 1$ then $f(1) = \ln(1) = 0$ and since $f'(x) = \frac{1}{x}$, so $f'(1) = \frac{1}{(1)} = 1$. Therefore the linearization is:

$$L(x) = 1(x - 1) + 0 \Leftrightarrow L(x) = x - 1$$

Therefore we can say: $\ln(1.5) = f(1.5) \approx L(1.5) = 1.5 - 1 = 0.5$.

Definition: If $y = f(x)$ where f is a differentiable function, then

- the **differential** dx is an independent variable; that is, dx can be given the value of any real number.
- the **differential** dy is then defined by the equation $dy = f'(x)dx$.

So dy is a dependent variable that depends on the values of x and dx .

Note: This definition is in agreement with Leibniz notation for the derivative. In particular, $\frac{dy}{dx} = f'(x)$.

General Idea:

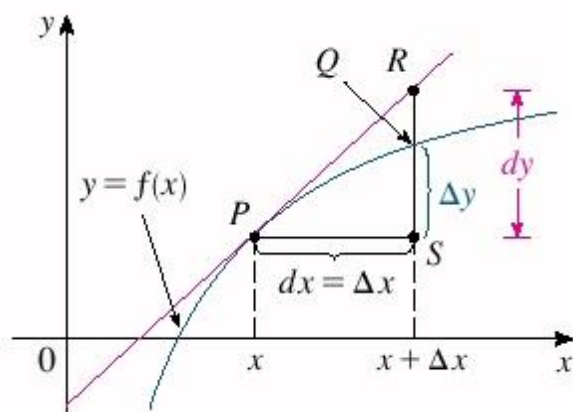
The geometric meaning of a differential can be seen in the following diagram.

Suppose $y = f(x)$ and P is a point on the graph of this function with coordinates: $(x, f(x))$.

Let Q be the point on the graph of this function with coordinates $(x + \Delta x, f(x + \Delta x))$.

Then, we think of dx as Δx (the change in x). Mathematically, $dx = \Delta x$.

In chapter 2 we defined the corresponding change in y as $\Delta y = f(x + \Delta x) - f(x)$. Note that $dy \neq \Delta y$ in general since $dy = f'(x)dx = f'(x)\Delta x$. However, we can think of dy and Δy as approximations of each other. The diagram below illustrates this.



$\Delta x = dx$ both represents the change in the value of x .

Δy represents the change in the y -coordinate of $f(x)$.

dy represents the change in the y -coordinate of the tangent line.

Note: Very importantly we see from the diagram above that the linearization does a nice job of approximating the function at $x + \Delta x$ when dy and Δy are close to each other. In general, dy and Δy get closer to each other as Δx gets smaller.

Example 3: Suppose that $f(x) = x^3 - 1$. Find the values of Δy and dy (rounded to 4 decimals) if x changes:

- a) from 2 to 2.4.

We first note that $\Delta x = 2.4 - 2 = 0.4$. Then by definition:

$$\Delta y = f(2.4) - f(2) = 5.8424 \text{ and}$$

$$dy = f'(x)dx \Rightarrow dy = 3x^2 \cdot dx \Rightarrow dy = 3(2)^2 \cdot (0.4) = 4.8$$

- b) from 2 to 2.05

We first note that $\Delta x = 2.05 - 2 = 0.05$. Then by definition:

$$\Delta y = f(2.05) - f(2) \approx 0.6151 \text{ and}$$

$$dy = f'(x)dx \Rightarrow dy = 3x^2 \cdot dx \Rightarrow dy = 3(2)^2 \cdot (0.05) = 0.6$$

Note: This example is in line with our intuition that dy and Δy get closer to each other as Δx gets smaller.

Note: The differential notation introduced above will be common in upcoming chapters. It will also continue to be utilized in second semester calculus, and it will be generalized in third semester calculus.

Example 4: Let $y = \sqrt{x-3}$.

- a) Find the values of Δy and dy (rounded to 4 decimals) if x changes from 4 to 7

We first note that $\Delta x = 7 - 4 = 3$. Then by definition:

$$\Delta y = f(7) - f(4) = \sqrt{4} - \sqrt{1} = 1 \text{ and}$$

$$dy = f'(x)dx \Rightarrow dy = \frac{1}{2\sqrt{x-3}} \cdot dx \Rightarrow dy = \frac{1}{2\sqrt{1}} \cdot (3) = \frac{3}{2} = 1.5$$

- b) Make a sketch of $y = \sqrt{x-3}$, showing the values of Δy and dy .

